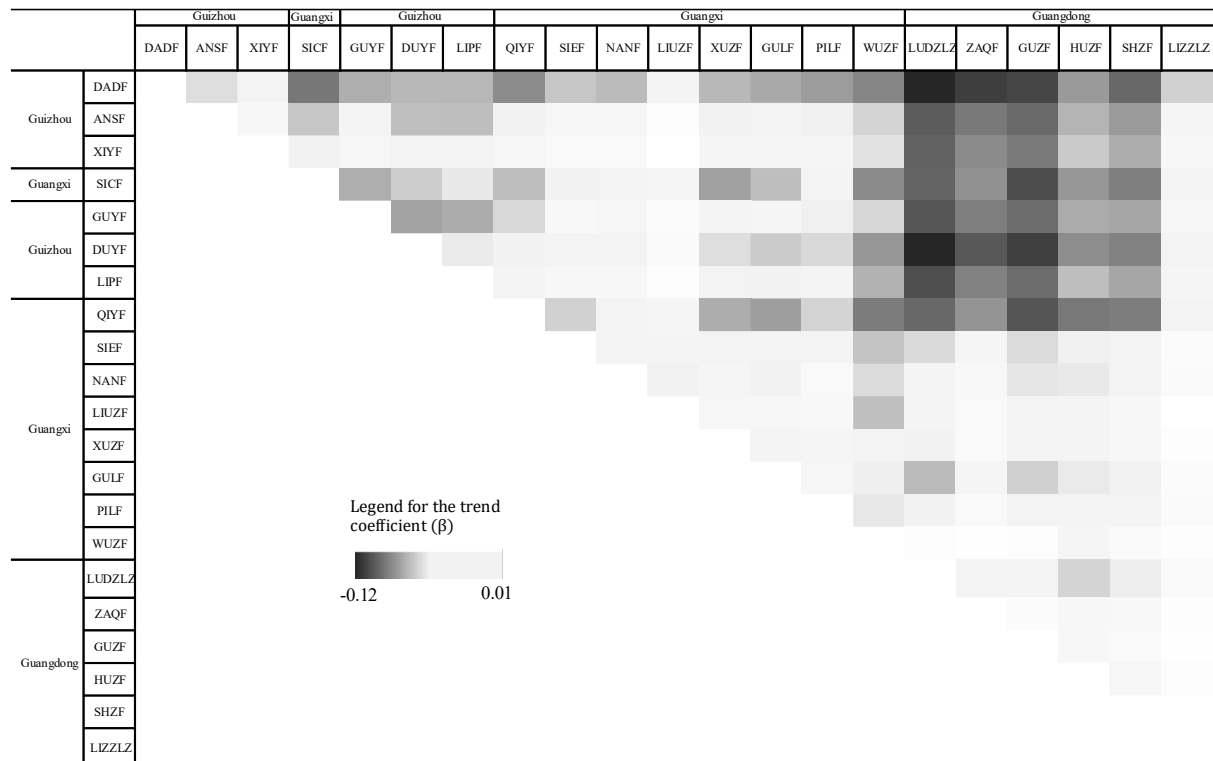
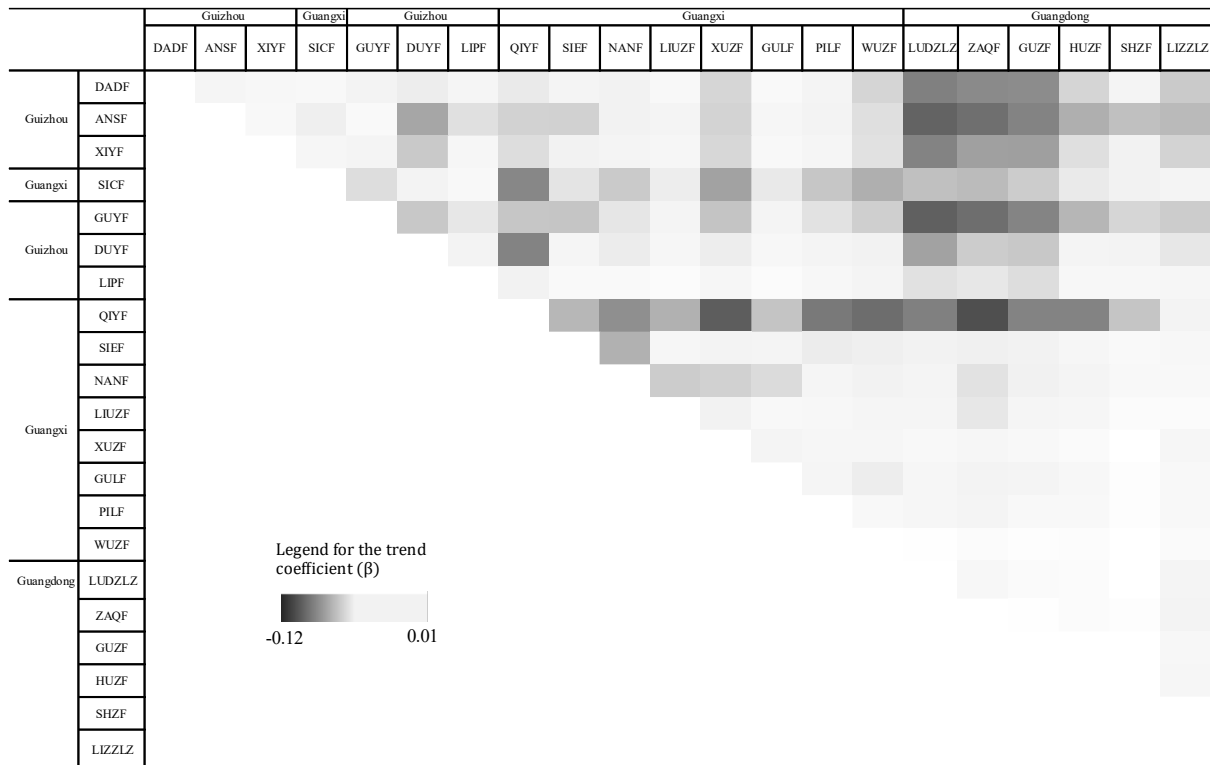


Appendix 1: Complete heatmaps for the highest and lowest prices of the mid-grade rice



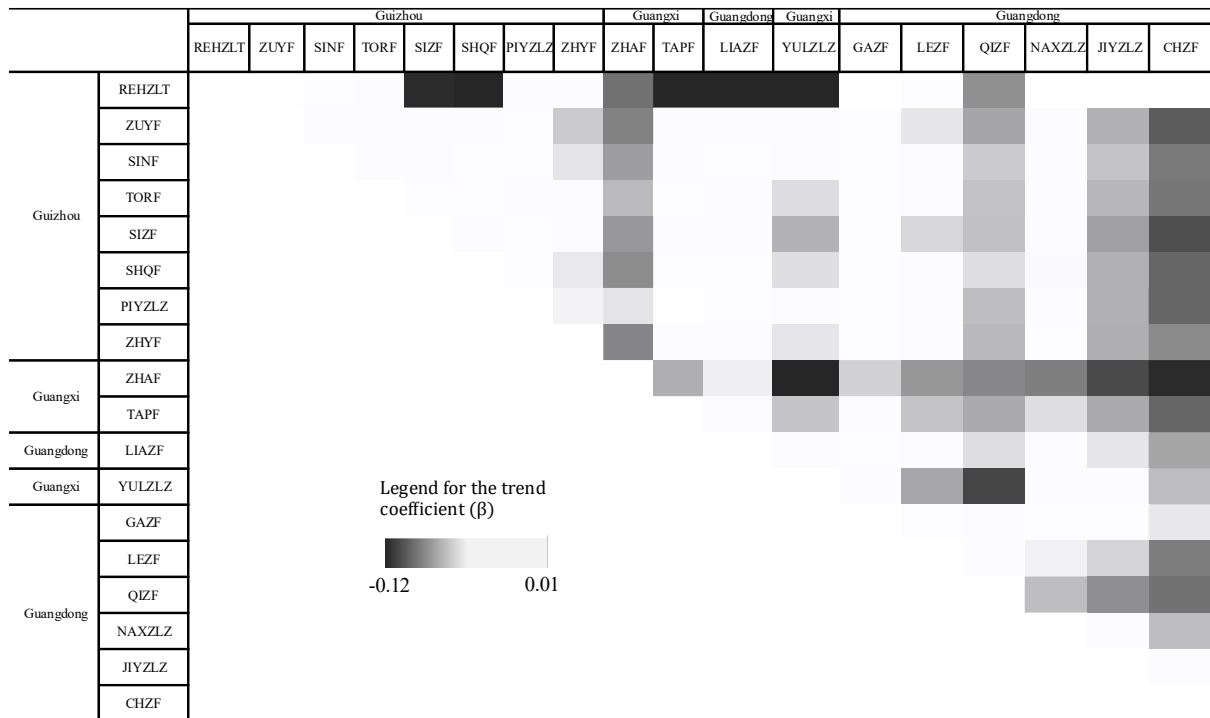
Note: DA DF: Da ding Fu (大定府), AN SF: An shun Fu (安顺府), XIY F: Xing yi Fu (興義府), SICF: Si cheng Fu (泗城府), GU YF: Gui yang Fu (貴陽府), DU YF: Du yun Fu (都勻府), LIPE: Li ping Fu (黎平府), QIY F: Qing yun Fu (慶遠府), SIEF: Si en Fu (思恩府), NANF: Na ming Fu (南寧府), LIUZF: Liu zhou Fu (柳州府), XU ZF: Xu zhou Fu (潯州府), GU LF: Gui lin Fu (桂林府), PILF: Ping le Fu (平樂府), WUZF: Wu zhou Fu (梧州府), LUD ZLZ: Lu ding zhi li zhou (羅定直隸州), ZAQ F: Zhao qing Fu (肇慶府), GU ZF: Guang zhou Fu (廣州府), HU ZF: Hu zhou Fu (惠州府), SHZF: Shao zhou Fu (韶州府), LIZZLZ: Lian zhou Zhi li zhou (連州直隸州).

HEAT MAP 1 Prefectures within the Pearl River System, highest prices of mid-grade rice. The upper-right quadrant, representing pairings between prefectures in Guizhou and Guangdong, displays the most pronounced shading, indicating that the trend toward market integration is strongest. This result suggests that the market integration effect among prefectures connected through the Pearl River system becomes more pronounced as distance increases. In contrast, neighbouring provinces – such as prefectural pairings between Guizhou and Guangxi, Guangxi and Guangdong, and within individual provinces – exhibit less pronounced integration effects than those observed between Guizhou and Guangdong.



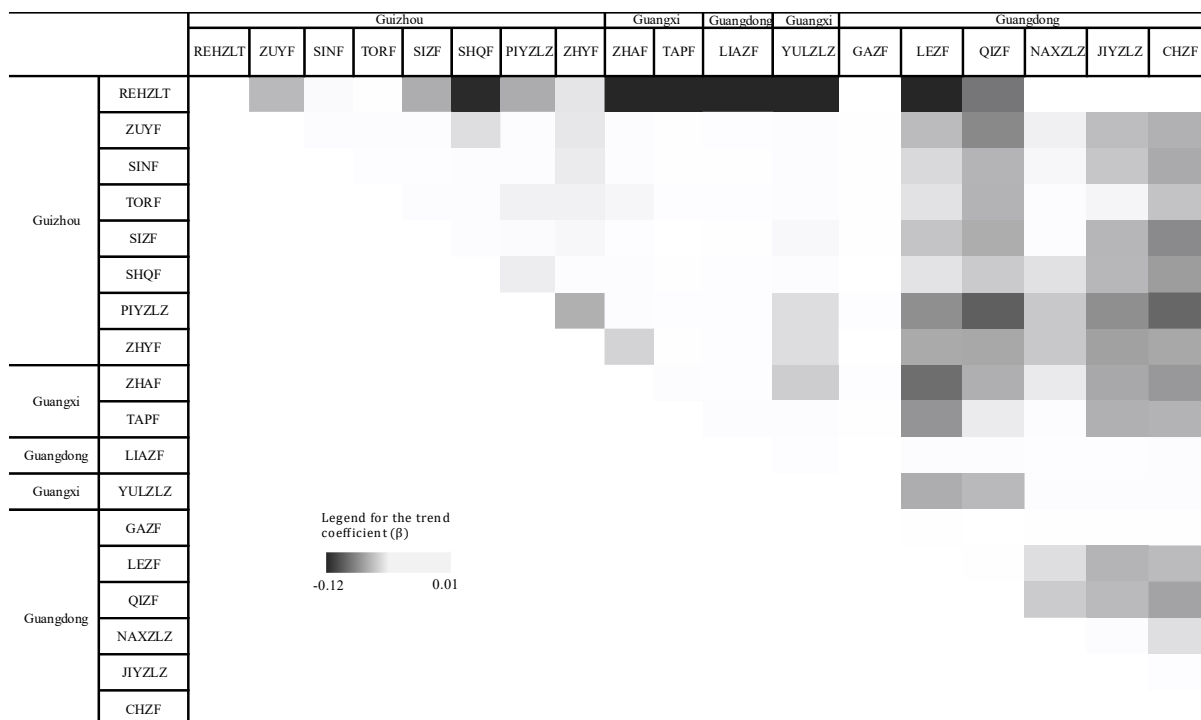
HEAT MAP 2 Prefectures within the Pearl River System, lowest prices of mid-grade rice.

Despite of the similar color distribution, the overall color tone of HEAT MAP 2 is brighter than HEAT MAP 1, suggests that the integration of mid-grade rice lowest prices was not significant as the one of the highest prices. In addition, compare to the other prefectures within Pearl River system, Qingyuan Fu and Sichen Fu, Guangxi province, relatively more strongly integrated with the prefectures of Guangdong and Guangxi provinces, the β values are in between -0.105 and -0.063.



Note: REHZLT: Renhuai Zhiliting (仁懷直隸廳), ZUYF: Zunyi Fu (遵義府), SINF: Siman Fu (思南府), TORF: Tongren Fu (銅仁府), SIZF: Sizhou Fu (思州府), SHQF: Shiqian Fu (石阡府), PIYZLZ: Pingyue Zhilizhou (平越直隸州), ZHYF: Zhenyuan Fu (鎮遠府), ZHAF: Zhen'an Fu (鎮安府), TAPF: Taiping Fu (太平府), LIAZF: Lianzhou Fu (廉州府), YULZLZ: Yulin Zhilizhou (郁林直隸州), GAZF: Gaohou Fu (高州府), LEZF: Leizhou Fu (雷州府), QIZF: Qiongzhou Fu (瓊州府), NAXZLZ: Nankong Zhilizhou (南甯直隸州), JIYZLZ: Jiaying Zhilizhou (嘉應直隸州), CHZF: Chaozhou Fu (潮州府).

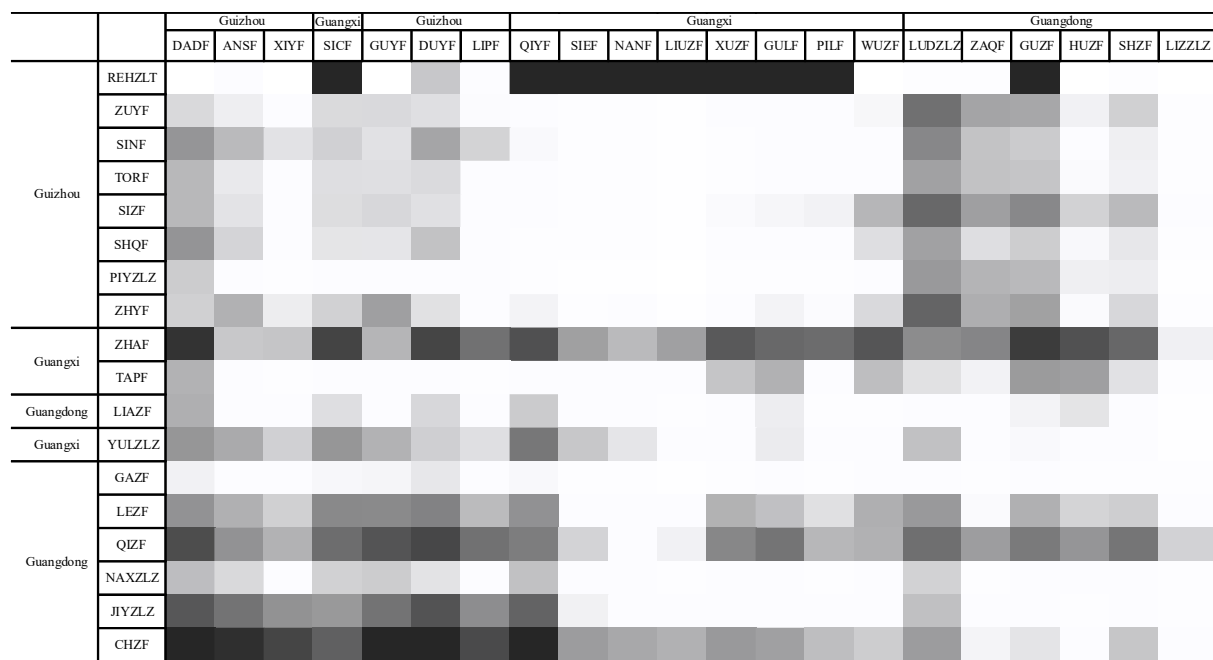
HEAT MAP 3 Prefectures outside the Pearl River System, highest prices of mid-grade rice. The color distribution reveals several patterns of regional integration. (1) First, Renhuai Zhiliting in Guizhou, located at the uppermost reaches, exhibits particularly strong integration with Shiqian Fu and Sizhou Fu in Guizhou; Zhen'an Fu, Taiping Fu and Yulin Zhilizhou in Guangxi; and Lianzhou Fu and Qiongzhou Fu in Guangdong. (2) Zhen'an Fu and Yulin Ting in Guangxi, and Qiongzhou Fu, Chaozhou Fu, and Jiaying Ting in Guangdong display generally strong integration trends with other non-river-system prefectures across the three provinces. (3) The upper-left region, presenting pairs of prefectures within Guizhou, are almost white, implying relatively weak integration trends.



Note: REHZLT: Renhuai Zhiqing (仁懷直隸廳), ZUYF: Zunyi Fu (遵義府), SINF: Siman Fu (思南府), TORF: Tongren Fu (銅仁府), SIZF: Sizhou Fu (思州府), SHQF: Shiqian Fu (石阡府), PIYZLZ: Pingyue Zhiqing (平越直隸州), ZHYF: Zhenyuan Fu (鎮遠府), ZHAF: Zhen'an Fu (鎮安府), TAPF: Taiping Fu (太平府), LIAZF: Lianzhou Fu (廉州府), YULZLZ: Yulin Zhiqing (郁林直隸州), GAZF: Gaoguo Fu (高州府), LEZF: Leizhou Fu (雷州府), QIZF: Qiongzhou Fu (瓊州府), NAXZLZ: Nanxiong Zhiqing (南雄直隸州), JIYZLZ: Jiaying Zhiqing (嘉應直隸州), CHZF: Chaozhou Fu (潮州府).

HEAT MAP 4 Prefectures outside the Pearl River System, lowest prices of mid-grade rice.

The color mapping of the mid-grade rice lowest prices of prefecture pairs outside the Pearl River System is similar to the one of the highest prices, implies the integrating patterns of highest and lowest prices are alike. Yet the integrations of Zhen'an Fu, Guangxi province, with other prefectures are less significant than the ones of the highest prices.

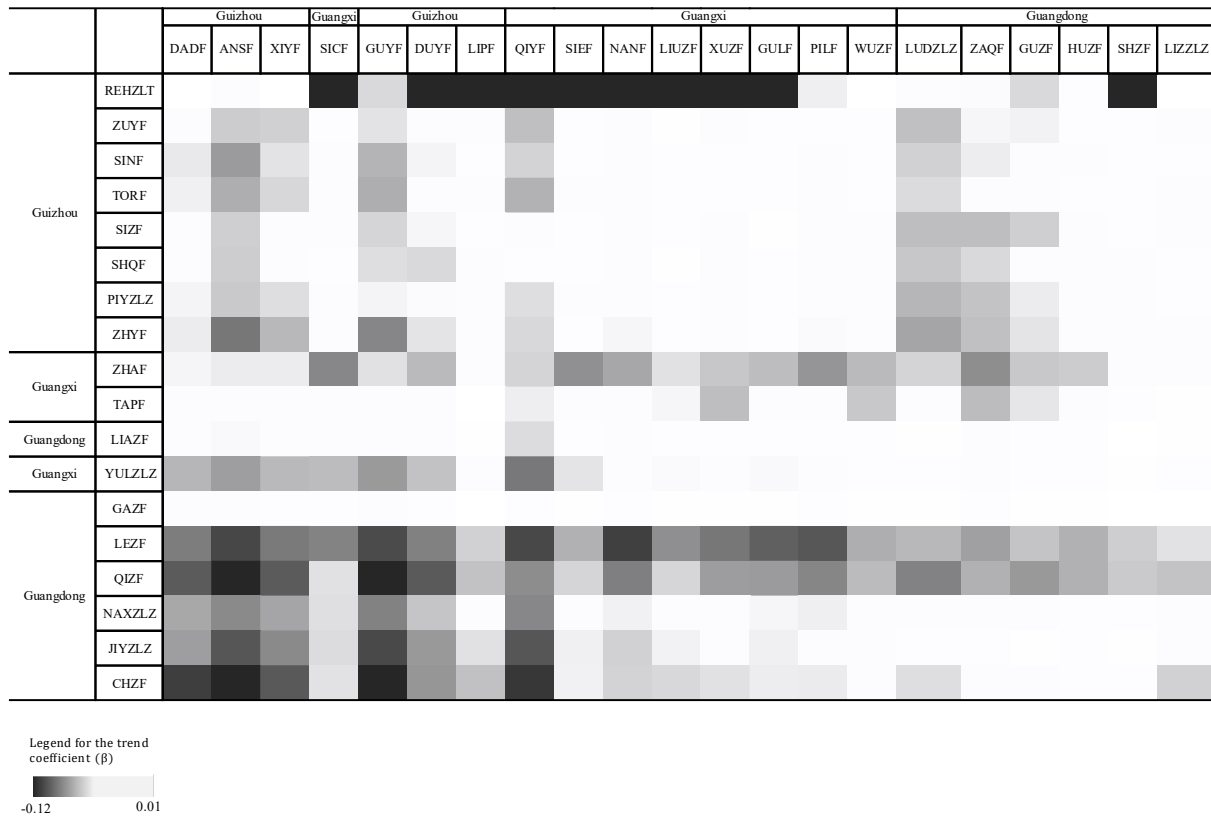


Legend for the trend coefficient (β)



Note: DA DF: Dàdìng Fú (大定府), AN SF: Ànshùn Fú (安順府), XIY F: Xīngyì Fú (興義府), SICF: Sìchéng Fú (泗城府), GU YF: Guìyáng Fú (貴陽府), DU YF: Dūyún Fú (都勻府), LIPF: Lǐpíng Fú (黎平府), QIY F: Qīngyuán Fú (慶遠府), SIEF: Sīēn Fú (思恩府), NA NF: Nánmíng Fú (南寧府), LIUZF: Liúzhōu Fú (柳州府), XU ZF: Xúzhōu Fú (潯州府), GU LF: Guìlín Fú (桂林府), PILF: Pínglè Fú (平樂府), WUZF: Wúzhōu Fú (梧州府), LUDZLZ: Luòdìng Zhìlìzhōu (羅定直隸州), ZAQF: Zhāoqīng Fú (肇慶府), GU ZF: Guāngzhōu Fú (廣州府), HU ZF: Huìzhōu Fú (惠州府), SHZF: Shào zhōu Fú (韶州府), LIZZLZ: Liánzhōu Zhìlìzhōu (連州直隸州), REHFLT: Rénhé zhìlìzhōu (仁懷直隸廳), ZUY F: Zūnyì Fú (遵義府), SINF: Sīmǎn Fú (思南府), TORF: Tóngān Fú (銅仁府), SIZF: Sīzhōu Fú (思州府), SHQF: Shíqián Fú (石阡府), PIYZLZ: Píngyuè Zhìlìzhōu (平越直隸州), ZHY F: Zhēnyuán Fú (鎮遠府), ZHA F: Zhēn'ān Fú (鎮安府), TAPF: Tàipíng Fú (太平府), LIAZF: Liánzhōu Fú (廉州府), YU LZLZ: Yùlín Zhìlìzhōu (郁林直隸州), GAZF: Gāozhōu Fú (高州府), LEZF: Lézōu Fú (雷州府), QIZF: Qióngzhōu Fú (瓊州府), NA XZLZ: Nánxīng Zhìlìzhōu (南雄直隸州), JIY ZIZ: Jiāyíng Zhìlìzhōu (嘉應直隸州), CHZF: Cháo zhōu Fú (潮州府).

HEAT MAP 5 Interactions between Pearl River System and non-river-system prefectures, highest prices of mid-grade rice. Several distinct spatial patterns of integration trends. (1) The upper-right and lower-left quadrants display relatively darker shading, indicating stronger integration trends. The former quadrant represents pairings between river-system prefectures in Guangdong and non-river-system prefectures in Guizhou, and the later quadrant represents pairings between river-system prefectures in Guizhou and non-river-system prefectures in Guangdong. (2) The upper-left and lower-right quadrants represent, respectively, within-province integration between river-system and non-river-system prefectures in Guizhou and Guangdong. Although the shading is generally lighter than in the upper-right and lower-left quadrants, a substantial number of pairings still display noticeable color intensity, implying a considerable degree of integration between river-system and non-river-system prefectures. (3) From the perspective of river-system prefectures in Guangxi, a conspicuous white region appears in the upper-middle portion of the heat map, indicating that integration between Guangxi river-system prefectures and non-river-system prefectures in Guizhou is very weak. By contrast, integration between Guangxi river-system prefectures and non-river-system prefectures in Guangdong is much more pronounced.



Note: DADF: DadingFu (大定府), ANSF: AnshunFu (安顺府), XIYF: XingyiFu (興義府), SICF: SichangFu (泗城府), GUYF: GuiyangFu (貴陽府), DUYF: DuyunFu (都勻府), LIPF: LipingFu (黎平府), QIYF: QingyuanFu (慶遠府), SIEF: Si'enFu (思恩府), NAF: NanmingFu (南寧府), LIUF: LiuzhouFu (柳州府), XUZF: XunzhouFu (潯州府), GULF: GuilinFu (桂林府), PILF: PinglingFu (平樂府), WUZF: WuzhouFu (梧州府), LUDZLZ: LuodingZhiliZhao (羅定直隸州), ZAQF: ZhaoqingFu (肇慶府), GUZF: GuangzhouFu (廣州府), HUF: HuizhouFu (惠州府), SHZF: ShaohuFu (韶州府), LIZLZ: LianzhouZhiliZhao (連州直隸州), REHZLT: RenhuaiZhiliZhao (仁懷直隸州), ZUYF: ZunyiFu (遵義府), SINF: SimanFu (思南府), TORF: TongrenFu (銅仁府), SIZF: SizhouFu (思州府), SHQF: ShiqianFu (石阡府), PIYZLZ: PingyueZhiliZhao (平越直隸州), ZHYF: ZhenyuanFu (鎮遠府), ZHAF: Zhen'anFu (鎮安府), TAPF: TaipingFu (太平府), LIAZF: LianzhouFu (廉州府), YULZLZ: YulinZhiliZhao (郁林直隸州), GAZF: GaizhouFu (高州府), LEZF: LeizhouFu (雷州府), QIZF: QiongzhouFu (瓊州府), NAXZLZ: NaxiongZhiliZhao (南雄直隸州), JYZLZ: JiayingZhiliZhao (嘉應直隸州), CHZF: ChaozhouFu (潮州府).

HEAT MAP 6 Interactions between Pearl River System and non-river-system prefectures, lowest prices of mid-grade rice. In overall, the color of heat map 6 is lighter than heat map 5, indicating the less strong integration of lowest prices. But the pattern of two heat maps are similar. Among the portions of the heat map 6, the lower-left region is the darkest, suggests that the integration of the river-system prefectures in Guizhou with the non-river-system ones in Guangdong is visibly stronger than other regions of this heatmap .